

HONOLULU PIER 4, HI, USA

WMO: 994007

Lat: 21.3033N

Lon: 157.8645W

Elev: 10

StdP: 14.69

Time Zone: -10.00 (W10)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	66.5	67.9	N/A	N/A	N/A	N/A	N/A	N/A	15.6	73.0	13.7	73.8	3.4	60	0.253

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	7.3	86.0	N/A	85.1	N/A	84.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.0	60

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		Extreme Max WB	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
12.9	11.6	10.5	63.4	86.7	1.1	1.5	62.6	87.8	61.9	88.6	61.3	89.5	60.5	90.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	77.1	73.7	73.5	73.5	75.4	77.0	78.5	79.6	80.4	80.4	79.5	77.4	75.5
	DBStd	3.11	1.84	2.26	2.16	2.00	1.72	1.28	1.60	1.54	1.59	1.53	1.78	1.72
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9878	734	659	728	763	836	855	919	941	913	915	823	792
	CDD65	4403	269	239	263	313	371	405	454	476	463	450	373	327
	CDH74	28928	730	631	623	1301	2147	3025	3926	4475	4371	3889	2409	1401
	CDH80	3469	2	7	5	31	122	287	600	841	858	545	146	25
Wind	WSAvg	4.9	4.7	5.0	5.3	5.2	5.1	5.1	5.0	5.0	4.7	4.4	4.8	4.6
Precipitation	PrecAvg	18.8	2.7	2.0	2.0	1.1	1.0	0.4	0.6	0.6	0.8	1.8	2.4	3.0
	PrecMax	42.8	12.8	9.4	8.4	8.9	7.2	2.4	2.7	7.6	4.5	11.1	14.7	17.3
	PrecMin	3.7	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0
	PrecStd	8.6	2.9	2.2	1.9	1.5	1.3	0.6	0.6	1.1	0.8	2.0	2.9	3.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.2	81.1	80.6	82.2	83.8	84.3	86.3	87.0	86.9	86.1	84.2	82.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	78.9	79.3	79.6	81.0	82.3	83.3	85.0	85.8	86.0	84.8	82.7	80.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	78.1	78.2	78.3	80.1	81.4	82.5	84.1	84.7	85.0	84.0	81.8	79.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	77.3	77.3	77.2	79.2	80.5	81.8	83.2	83.8	84.0	82.9	80.8	78.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	7.0	6.9	6.8	6.9	7.0	7.1	7.0	7.1	7.3	7.0	6.6	6.9
		MCDBR	7.5	7.7	7.6	7.6	7.7	7.6	7.4	7.6	7.7	7.7	7.3	7.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.366	0.373	0.381	0.391	0.384	0.371	0.369	0.368	0.361	0.380	0.379	0.361	0.361
	taud	2.567	2.535	2.521	2.482	2.533	2.577	2.591	2.594	2.617	2.568	2.562	2.602	2.602
	Ebn at Noon	288	292	293	290	288	290	290	292	295	286	282	286	286
	Edh at Noon	30	32	34	36	34	32	31	31	30	31	30	28	28
All-Sky Solar Radiation	RadAvg	1409	1586	1816	2050	2194	2297	2259	2171	2030	1707	1440	1302	1302
	RadStd	61	112	190	117	124	65	47	86	61	53	37	94	94

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+122

Nomenclature: See separate page